Savitribai Phule Pune University Data Structures and Algorithms (210252) SE Computer Engineering 2019 Pattern

Total Marks: 70 | Time: 2½ Hours

Instructions:

1. Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
2. Neat diagrams must be drawn wherever necessary.
3. Figures to the right indicate full marks.
4. Assume suitable data if necessary.

Unit III --- Graphs (18 Marks)

Q1) a) Write an algorithm for depth-first traversal of a graph. b) Construct the minimum spanning tree (MST) for the given graph using Prim's algorithm starting from vertex 6. [6]

```
      1 ---2--- 2
     /|         |\
   4 |         | 3
     \|         |/
      3---5--- 4
       \      /
        \   /
         \ /
          5
```

(Edge weights are marked on edges) c) What is topological sorting? Find the topological sorting of the given graph. [6]

```
     A ---> B ---> D
     |      |      |
     v      v      v
     C ---> E ---> F
```

OR [6]

Q2) a) Write an algorithm for breadth-first traversal of a graph. b) Using Prim's algorithm, find the cost of the minimum spanning tree (MST) of the given graph starting from vertex a. [6]

```
     a ---3--- b ---1--- c
     |         |         |
     4         2         5
     |         |         |
     d ---6--- e ---2--- f
```

c) Define the following terms: i) Complete graph ii) Connected graph iii) Subgraph [6] [6]

Unit IV --- Search Trees (17 Marks)

Q3) a) Construct an AVL tree by inserting numbers from 1 to 8. Show all rotations. b) Define a Red-Black tree. List its properties. Give an example. c) Write functions for RR and RL rotations with respect to an AVL tree. [6] [6] [5]

OR

Q4) a) Construct an AVL tree for the following data: 50, 25, 10, 5, 7, 3, 30, 20, 8, 15. Show all intermediate steps and rotations. b) Explain with an example the K-dimensional tree (K-D tree). [6]
c) Explain static and dynamic tree tables with suitable examples. [5] [6]

Unit V --- Indexing and Multiway Trees (18 Marks)

Q5) a) Construct a B-Tree of order 3 by inserting numbers from 1 to 10. Show all intermediate steps.
b) Explain primary index, secondary index, sparse index, and dense index with examples. [9] [9]

OR

Q6) a) Construct a B-Tree of order 5 with the following data: D, H, Z, K, B, P, Q, E, A, S, W, T, C, L, N, Y, M. Show all intermediate steps. b) What is a trie tree? Explain insert and search operations on it with an example. [9]
[9]

Unit VI --- File Organization (17 Marks)

Q7) a) Explain multilist files and coral rings. b) What is sequential and indexed sequential file organization? State their advantages and disadvantages. [9]
[8]

OR

Q8) a) Explain inverted files and cellular partitions. b) Explain direct access file organization. State its advantages and disadvantages. [9]
[8]